Chapter 23. Future of SRE

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Throughout this book, we've explored a number of transitions. Transitions are nothing new—Google specifically, and the technology industry generally, have embraced and thrived through many of them. Frequently, the ultimate outcome looked very different from what we thought it would at the outset. With this in mind, we look into the future of SRE with a sense of humility. It's not possible to fully predict what will come, but we can make an informed guess about what's not likely to change about the SRE practice and what we suspect definitely will.

What Won't Change

First, reliability remains as important as ever. In fact, the importance of reliability is *increasing* as the world becomes ever more reliant on technology. The products and infrastructure provided by Google and other innovative organizations are becoming increasingly critical to the daily lives of billions of users and to organizations globally. This means that our infrastructure is becoming *societal infrastructure* to governments, nongovernmental organizations, companies, and people around the world. We don't take this responsibility lightly—it is central to our culture and our intention that "Google just works." If we project forward, it is reasonable to assume that this global reliance will continue to increase (and perhaps dramatically), yet expectations will remain the same. As complexity grows, our approaches to reliability will change, but our mission will not.

Second, we fully expect that humans will remain involved in the practice of SRE and the broader work of delivering reliable systems and products. As AI capabilities expand at pace, we view stewardship of our production systems as a collaboration between humans and AI. At present, our human SREs continue to perform most production changes, often with an AI companion. We expect that over time, the division of work between humans and AI will shift, with AI taking on more of the investigation, detection, mitigation, and post-incident response. At the same time, our most complex problems are often novel, don't follow previously understood patterns, and are best solved by humans. We expect that will be the case for the foreseeable future. As this book has shown, many as-

pects of SRE go beyond production management and incident response, and human judgment continues to be essential there as well.

What Is Transforming

Transitions involve transformations of many kinds, from process to platform to culture. We are hitting a major inflection point in SRE, and we need to rethink how we scale production while also striving to keep things simple. From abstracting away complexity to embracing agentic workflows, we are focused on making reliability the easy and default choice for our teams.

Shift Left Turns Into Shift Down

The desire to "shift left" elements of the product lifecycle has generated considerable discussion and led to many valuable outcomes. Shifting left ensures that domains such as quality, reliability, and security are addressed earlier in the product lifecycle, where issues can be resolved faster and at lower cost. That said, shifting left can also add considerable complexity and cognitive load to software development teams, which must now be experts not only in software development but also in quality, reliability, and security. This quickly becomes intractable even at small scale, and more so at the scale of large software companies.

This is why we have invested in platforms that allow us to "shift down" rather than "shift left" (see "Platform Engineering" and Richard Seroter's blog post on shifting down). Shifting down focuses on offloading these infrastructure and operational burdens onto the underlying platform. Instead of requiring developers to be full-stack experts who manage everything from Kubernetes configurations to network policies, organizations are encouraged to build or use platforms that abstract these complexities away, allowing developers to focus primarily on writing application code.

At its core, this strategy leverages managed services and opinionated frameworks that have best practices baked in by default. Converging on common runtime frameworks and platforms for rollouts, observability, and capacity management means the platform can handle the heavy lifting of scaling, availability, and compliance. By treating platforms as internal products that support engineers rather than new concepts they must master, we foster a more sustainable and innovative engineering culture while also making compliance with our production principles the easy choice (see "Google's Principles for Reliability and Resilience").

Risk as a Leading Indicator of Reliability

Traditionally, SRE has focused on availability as a key reliability indicator, forming the basis for many SLIs and SLOs. This hasn't always been the best approach, because availability is a trailing indicator of reliability: it is impacted when something goes wrong and the system is not available. However, the availability of a system is a manifestation of the inherent risk in the system and its dependencies. If we can identify that risk, we can start to understand the *leading* indicators of reliability rather than relying on the trailing ones.

Identifying risks can be done in a number of ways. For example, in the spirit of the continuous improvement culture of SRE, every incident—and every near miss—contains lessons that can alert us to risks in our systems. A good postmortem does not just identify the action items required to detect, mitigate, and prevent similar incidents in the future. A good postmortem should also include a catalog of the risks discovered.

There are also proactive ways to identify risk. Chapter 12 discusses STPA as a means for identifying systemic risks, and we expect application of this and similar methods to increase in the future. AI has the potential to identify risk through inspection of both design and architecture documents and source code, and we are seeing encouraging early results from this approach.

Once risks have been identified, they need to be cataloged and stored. We created a taxonomy, inspired by the Software Engineering Institute's taxonomy-based risk identification and approach, to classify them. Once we have a common classification scheme and a repository of risks, we can begin the process of systematically reducing them. The taxonomy is valuable in this context because it allows us to map impactful incidents to common categories, revealing recurring patterns and opportunities for improvement. For example, an analysis of this sort led us to conclude that many incidents were due to improper responses to overload, which encouraged us to build overload protections into our core RPC systems. This response to a widespread systemic risk made all Google systems better by default.

While identification and mitigation of common risks are largely manual processes today, AI will greatly accelerate both of them in the future.

Evolving the Practice of SRE with AI

It is difficult to overstate the impact that AI is already having on the practice of SRE. Chapter 21 discussed some of the ways we are employing AI to elevate our practice

across the hierarchy of production health. No other development is likely to have more impact than the introduction of AI agents into our production environments and the shift to agentic production.

Over the last year, we have witnessed a profound transition from dialog-based interactions with LLMs to agents. Agentic workflows have already transformed the software development process, and they will transform production management just as quickly. We are transitioning from a human-centric to a human-supervised model. It's not inconceivable that orders of magnitude more code will be created, resulting in orders of magnitude more production changes. To scale tools and processes to this new reality, we need to rethink everything, with agents at the center.

All this change marks an inflection point for SRE. To channel Marshall Goldsmith, "What got us here won't get us there."[1] What will? Well, to start:

New solutions for release volume and velocity

In the face of dramatic increases in the volume of code being produced, a key question emerges: how will we get all that code into production? If we adhere to our current release processes and velocity, the number of changes in each release will explode, increasing the risk of each release and making it more difficult to identify root causes and perform rollbacks. We must develop new, massively pipelined ways to release many smaller changes, which is a challenge at Google's scale.

Larger service landscape

Agentic software development promises not just that dramatically more code will be produced, but also that dramatically more systems will populate the overall service landscape. As the number of systems grows, so does the complexity of dependency management. AI also makes creating new tools far easier, meaning many teams will build tools to address their specific use cases. The resulting proliferation and potential for duplication will present a significant obstacle to SRE's goal of standardization. Innovation in dependency management, tool tracking, and standardization will be required to manage this increased complexity.

From specialist to generalist

While an SRE superpower is to see across product and systems boundaries when solving issues or architecting reliability, SREs remain specialized in certain domains. Specialization is required due to the scale and complexity of some of the domains, especially at the lower layers of our infrastructure. As we shift to agentic

production, SREs will have less direct knowledge of the systems being deployed, but their more general understanding of the principles of reliability will enable them to direct agents toward those goals.

SRE has a long history of innovation to change the practice of reliability engineering. In fact, the first edition of this book made it clear that a key responsibility for an SRE is to "automate themselves out of a job"—to apply technology and automation to eliminate manual and repetitive parts of our work. In a world where AI has the potential to transform the employment landscape across many industries, this mantra has broader implications.

We choose to take a view of the future that is based on abundance. As Demis Hassabis, CEO of Google DeepMind, recently stated: "Assuming we steward AGI safely and responsibly into the world, and we are trying to play our part in that, then we should enter an era of radical abundance. It would bring incredible productivity and prosperity for society and, perhaps for the first time in human history, we wouldn't view things in the world as zero-sum."[2]

So while it is important to consider the impact AI might have on SRE, and on society more generally, we choose to lean into a future of abundance. Rather than SREs automating themselves out of jobs, we believe that we will be automating ourselves *into* jobs. The SRE role will look different in the future, much as the role today looks different than it did when the first edition of this book came out. Still, many of the traits of a great SRE remain the same: strong curiosity, broad technical expertise, and great judgment. The future will include less toil, less on-call, and more autonomy. But what is really exciting is that the future of SRE will allow us to spend much more time on what SRE does best: reliability engineering at scale, enabling reliability by default, ensuring that our systems "just work."

1 Marshall Goldsmith and Mark Reiter, *What Got You Here Won't Get You There: How Successful People Become Even More Successful* (Hachette Books, 2007).

2 Steve Rose, "Demis Hassabis on Our AI Future: 'It'll Be 10 Times Bigger than the Industrial Revolution—and Maybe 10 Times Faster'", *The Guardian* (2025).

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